

$$Z=aX+bY,$$

$$\overset{X}{Y} \\ X \sim S(\alpha,0,1,0)$$

$$\varphi_X(t)=\exp(-|t|^\alpha).$$

$$depth(x,y)=1-F(x,y)$$

$$P\Big(aX+bY\geq au+bv\Big),$$

$$\overset{a,b}{(u,v)}$$

$$\overset{X}{Y} \\ Z=aX+bY$$

$$\varphi_Z(t)=\varphi_X(at)\varphi_X(bt)=\exp\Big[-\Big(|a|^\alpha+|b|^\alpha\Big)|t|^\alpha\Big].$$

$$\overset{Z}{Z} \sim S\Big(\alpha,0,c',0\Big),$$

$$c'=\left(|a|^\alpha+|b|^\alpha\right)^{1/\alpha}.$$

$$\frac{aX+bY}{c'}\sim S(\alpha,0,1,0).$$

$$\{aX+bY\geq au+bv\}$$

$$\left\{\frac{aX+bY}{c'}\geq\frac{au+bv}{c'}\right\}.$$

$$c=\frac{au+bv}{c'}.$$

$$s(a,b)=\frac{a\,u+b\,v}{\left(|a|^{\alpha}+|b|^{\alpha}\right)^{1/\alpha}},$$

$$\overset{a}{b} \\ \overset{(a,b)}{\{ (a,b) \in R^2 : \; |a|^{\alpha} + |b|^{\alpha} = 1 \}}.$$

$$s=au+bv,$$

$$\overset{(u,v)}{?}$$

$$\overset{?}{s} \\ f(t)=u\,t^{1/\alpha}+v\,(1-t)^{1/\alpha}, t\in[0,1],$$

$$a=t^{1/\alpha}, b=(1-t)^{1/\alpha},$$

$$\overset{a^\alpha+}{b^\alpha}=\\ \overset{1}{f(t)} \\ \overset{\alpha}{1}>\\ \overset{1}{f(t)} \\ t^*=\frac{1}{1+\left(\frac{v}{u}\right)^{\frac{\alpha}{\alpha-1}}}.$$

$$s_{\max}=\left(u^\beta+v^\beta\right)^{1/\beta}, \beta=\frac{\alpha}{\alpha-1}.$$

$$\overset{0}{1} \leqslant \\ \overset{1}{f(t)} \\ s_{\max}=\max\{u,v\}.$$

$$\overset{\alpha}{1} = \\ s_{\max}=\max\{u,v\},$$

$$X\sim S(\alpha,0,1,0)$$

$$F(x)=\frac{1}{2}-\frac{1}{\pi}\int_0^\infty\frac{Im\Big(e^{-itx}\varphi_X(t)\Big)}{t}\,dt.$$

$$\varphi_X(t)=\exp(-|t|^\alpha)$$

$$\frac{\alpha}{1}=\frac{s_{\max}}{\max\{u,v\}}\frac{\alpha}{2}=\frac{\alpha}{2}$$

$$\varphi_X(t)=\exp\Big(-t^2\Big),$$

$$\frac{\alpha}{1}> s_{\max}=\left(u^2+v^2\right)^{1/2},$$

$$\frac{\alpha}{1}>L^\beta=\frac{\alpha}{\frac{\alpha}{\alpha-1}}$$

$$\frac{\alpha}{1}\leq \frac{s_{\max}}{\max\{u,v\}}\frac{1}{aX+bY}$$

$$\frac{c'}{\left(|a|^\alpha+|b|^\alpha\right)^{1/\alpha}}c=\frac{au+bv}{c'},$$

$$s(a,b)=\frac{a\,u+b\,v}{\left(|a|^\alpha+|b|^\alpha\right)^{1/\alpha}},$$

$$\frac{s_{\max}}{\alpha}>\frac{1}{1}$$

$$s_{\max}=\left(u^\beta+v^\beta\right)^{1/\beta},\beta=\frac{\alpha}{\alpha-1}.$$

$$\frac{\alpha}{1}\leq$$

$$s_{\max}=\max\{u,v\}.$$

$$depth(u,v)=P\Big(aX+bY\geq au+bv\Big)=\frac{1}{2}-\frac{1}{\pi}\int_0^\infty\frac{\sin\Big(t\frac{au+bv}{c'}\Big)}{t}\exp(-t^\alpha)\,dt.$$

$$\frac{?}{1}=\frac{?}{2}\frac{\alpha}{\alpha}The depth function of population distribution in the Metrika$$